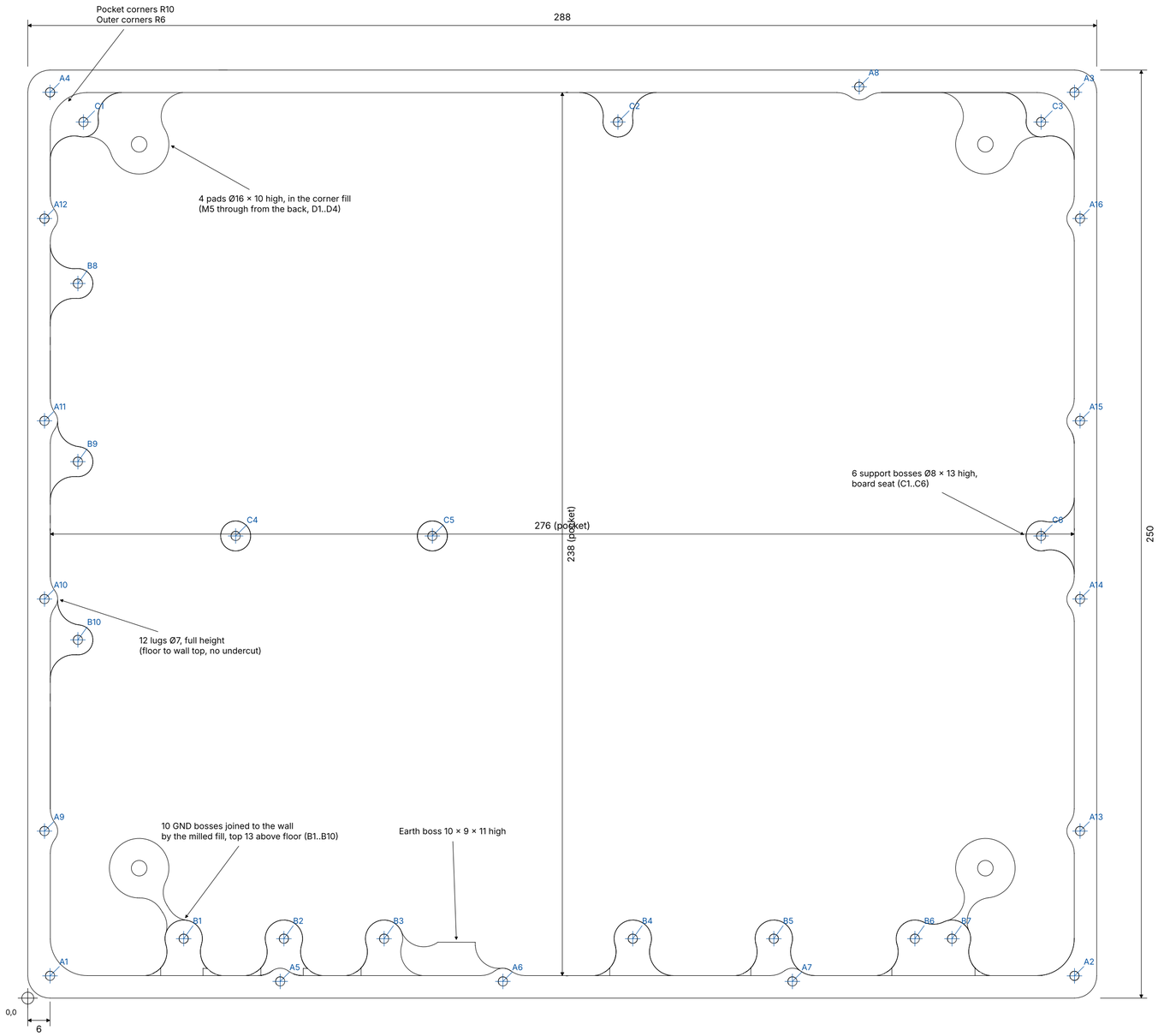
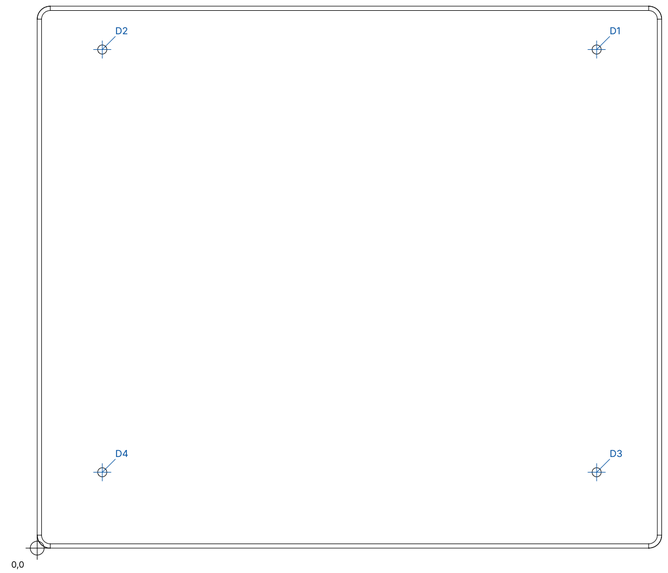


TRAY - PLAN (inside, seen from the lid side)

Floor 5 thick, walls 6, tray height 63. Tapped holes A, B, C: positions in the hole table (sheet 7), datum 0,0 = outer corner shown.

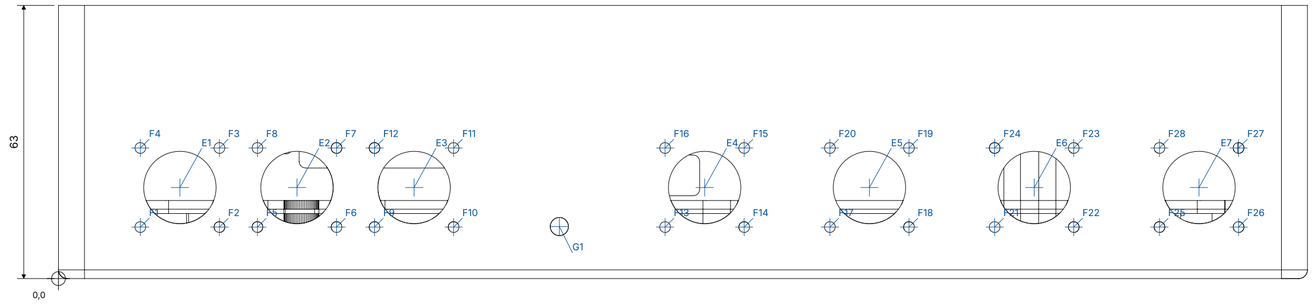


TRAY - BACK FACE (seen from the back plate)

Seen from the back: case x runs right to left. 4 × M5 (D1..D4) into the inside pads.

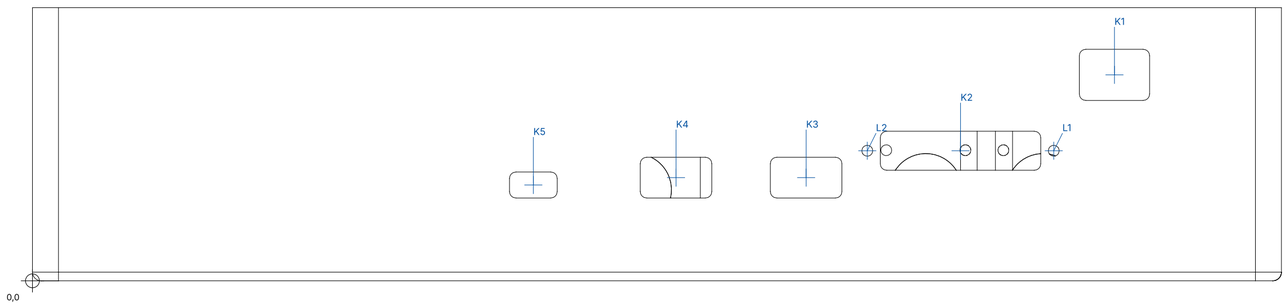
Back edge R2 all round. Flat face: mounts on the back plate via printed spacers.

Shack Relay Matrix rev D - RF unit case	
Tray: plan (inside) and back face	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 1 / 7 plan 1:1, back 1:2
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)



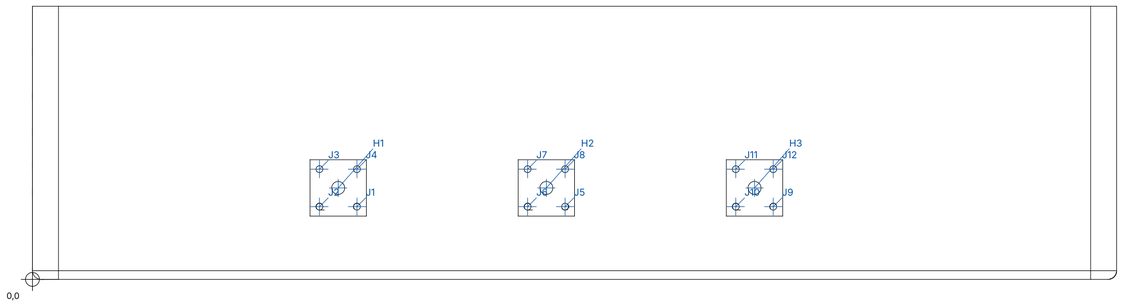
WALL 1 - SO-239 WALL (outside)

E: Ø16.7 THRU, flange outside, 4 × M3 on 18.24 square. G: earth strap M5 (tinned braid to the RVS earth bar).



WALL 2 - POWER / RJ45 WALL (outside)

K: cutouts THRU, corner R1.5, sizes in the hole table. EXP plate (sheet 5) covers K2 with 2 × M3 (L).



WALL 3 - SMA WALL (outside)

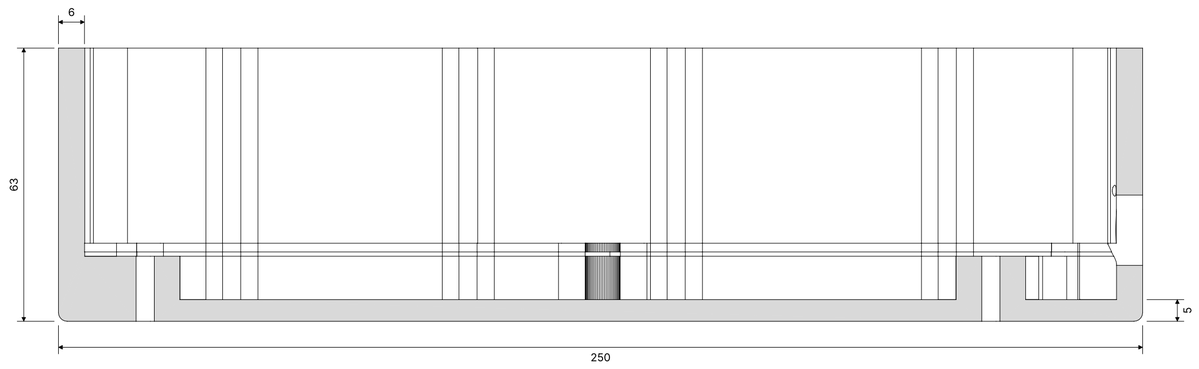
H: pocket 13 × 13 × 2 deep from outside (flange seat 12.7 square, corner R ≤ 1 allowed), Ø3.0 THRU at centre; J: 4 × M2 in the pocket floor.



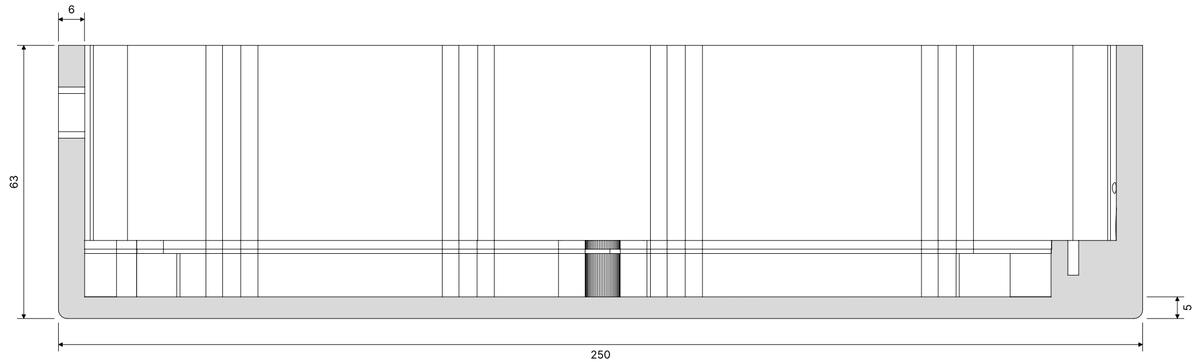
WALL 4 - PLAIN WALL (outside)

Wall 4 has no openings; the 4 side lugs are inside only.

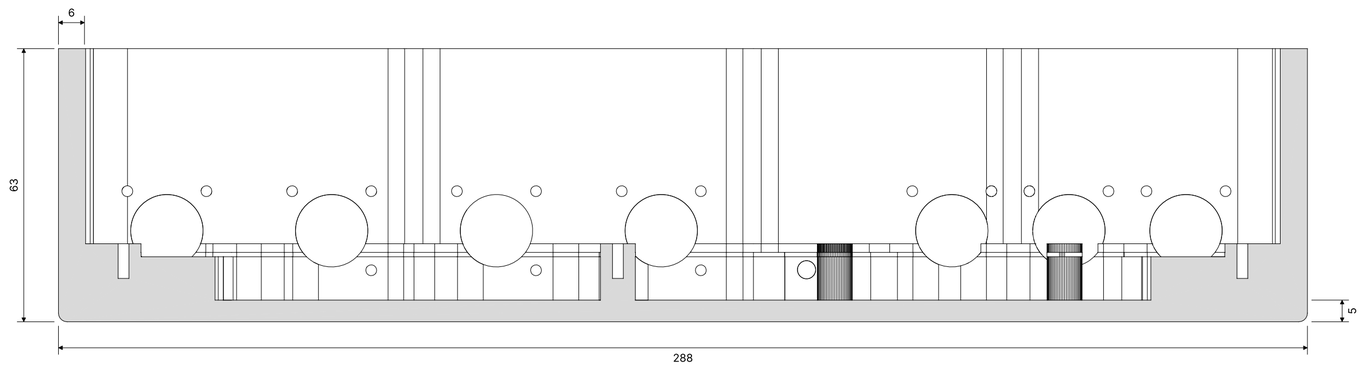
Shack Relay Matrix rev D - RF unit case	
Tray: the four walls, outside views	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 2 / 7 1:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)



SECTION A-A (x = 30: mounting pads, floor, walls)



SECTION B-B (x = 42: GND boss B1 joined to the SO-239 wall)



SECTION C-C (y = 236: support bosses S1..S3, lugs)

Section faces shaded.

Floor 5, walls 6, lugs Ø7 full height to the wall top.

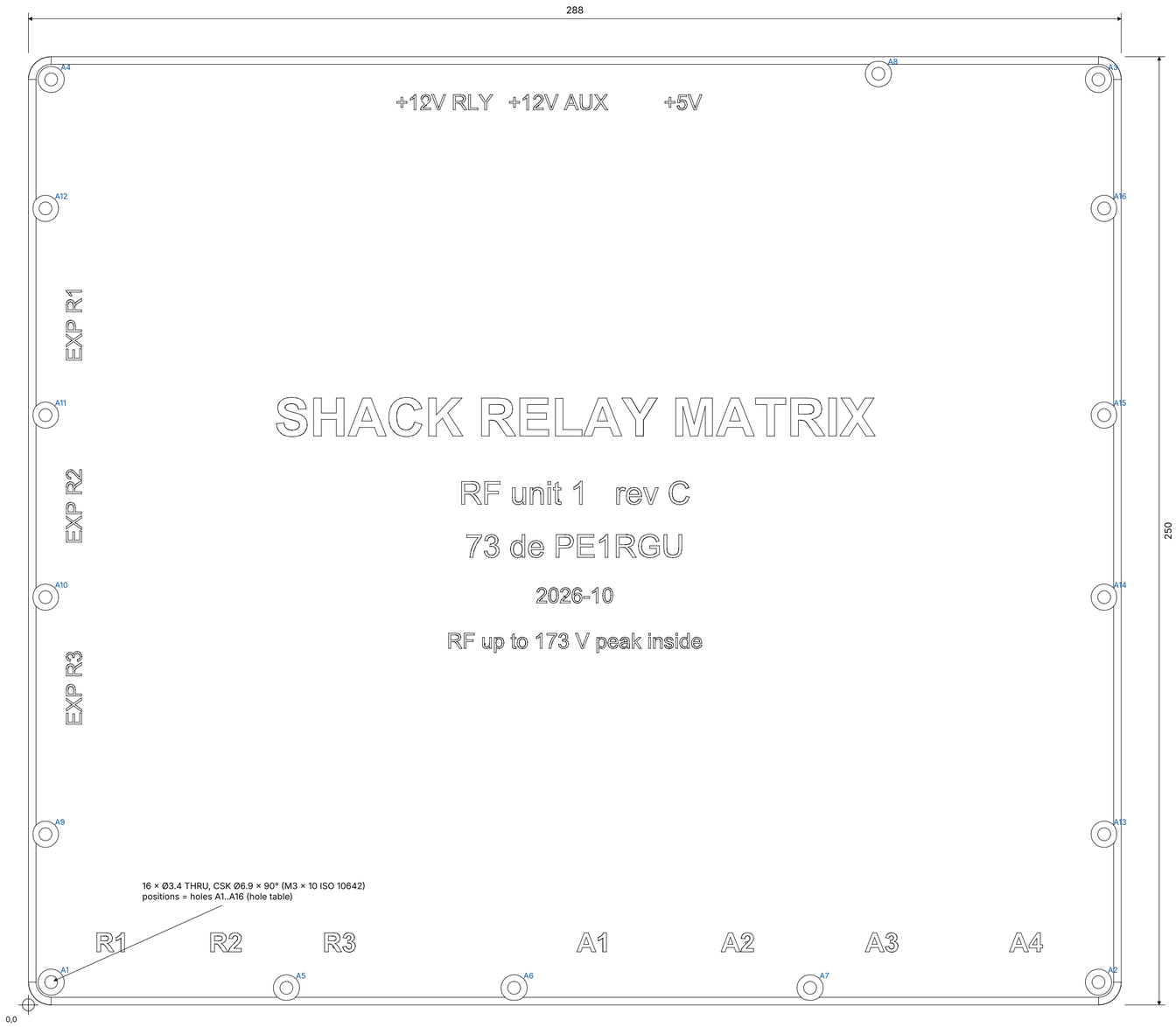
Mount pads 10 high (floor + pad = 15, M5 tapped through).

Support bosses and GND bosses 13 high (board seat).

Concave vertical edges in the pocket: R ≥ 6.5 (Ø12 cutter, D-204).

Floor edges sharp; R ≤ 1 from a corner-radius cutter accepted.

Shack Relay Matrix rev D - RF unit case	
Tray: sections	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 3 / 7 1:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)



LID - TOP (outside)

Lid 288 × 250 × 3, EN AW-6061-T6.

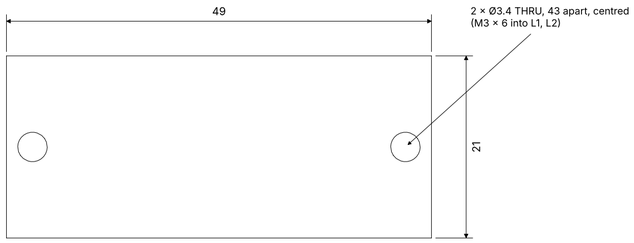
Outer corners R6, top edge R2, bottom edge sharp (seats on the tray).

Engraving 0.5 deep, Arial bold, as in rf-case.step (body Lid).

Optional: fill the engraving with white paint after coating.

Lid inside face and tray rim: no paint, conductive (RF bond).

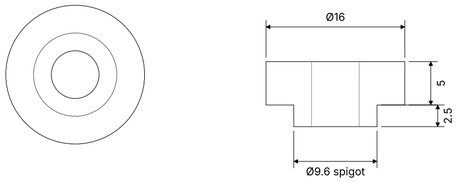
Shack Relay Matrix rev D - RF unit case	
Lid	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 4 / 7 1:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)



EXP PLATE (2 mm, EN AW-6061-T6)

Unit 1: blank. Unit 2: ribbon clamp and shield contact (same outline, slot to suit). Same finish as the tray.

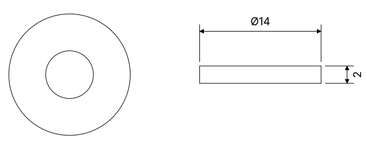
PRINTED SPACER (4×) 2:1



Bore Ø5.5 THRU. Spigot fits the Ø10 back-plate hole (plate 3 mm).

Flange side against the tray back, spigot into the plate.

PRINTED WASHER (4×) 2:1



Bore Ø5.5. Under the M5 × 16 head, behind the back plate.

Print: PETG or ASA (better PA-CF), 100 % infill, layers parallel to the flange. Plastic creeps under bolt load:

low torque, spring washer under the head, check the four M5 after the first summer. Stack from the back:

M5 × 16 A4 + spring washer + printed washer | back plate | printed spacer | tray. The tray is isolated from the plate;

the only bond is the tinned braid on G1 (earth M5 in wall 1) to the RVS earth bar.

Shack Relay Matrix rev D - RF unit case	
EXP plate and printed mounting parts	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 5 / 7 2:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)

(assembly render not found: [Errno 2] No such file or directory: '')

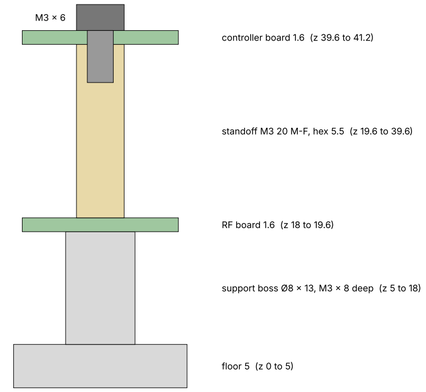
GND BOSS B1..B10 (7 SO-239 Ø10, 3 SMA Ø8), joined to the wall: RF board seated on the boss top (z 18), M3 × 6 from the board top.

LID: M3 × 10 countersunk ISO 10642 into A1..A16 (thread 8, lid 3, head flush with Ø6.9 countersink).

SO-239 83-798: flange outside wall 1, 4 x M3 × 6 into F (tapped through). SMA 901-9215: flange in pocket H, 4 x M2 × 4 into J (through).

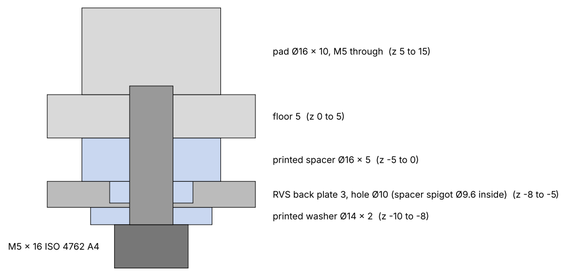
EXP plate: 2 x M3 × 6 into L (tapped through). Earth: M5 × 16 with braid lug into G1 (tapped through wall + boss, 15).

Heights: RF board underside z 18, controller underside z 39.6, DevKit top about z 57.2, lid inside z 63.



STACK AT S1, S2, S4, S5 (2:1)

S3, S6: no 20 mm standoff, M3 × 6 on the RF board (z 19.6) into the boss. Male stud: 6 mm through the board into the boss.



MOUNTING STACK D1..D4 (2:1)

Spring washer under the head. Tray isolated from the plate; bond only via G1.

Shack Relay Matrix rev D - RF unit case	
Assembly: stack-up, bosses and fasteners	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 6 / 7 details 2:1
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)

ID	Feature	Size	Depth	View	X	Y	Note
A1	Lid screw	M3	8	1 plan	6.00	6.00	tap drill 2.5, max 10 deep
A2	Lid screw	M3	8	1 plan	282.00	6.00	tap drill 2.5, max 10 deep
A3	Lid screw	M3	8	1 plan	282.00	244.00	tap drill 2.5, max 10 deep
A4	Lid screw	M3	8	1 plan	6.00	244.00	tap drill 2.5, max 10 deep
A5	Lid screw	M3	8	1 plan	68.00	4.50	tap drill 2.5, max 10 deep
A6	Lid screw	M3	8	1 plan	128.00	4.50	tap drill 2.5, max 10 deep
A7	Lid screw	M3	8	1 plan	206.00	4.50	tap drill 2.5, max 10 deep
A8	Lid screw	M3	8	1 plan	224.00	245.50	tap drill 2.5, max 10 deep
A9	Lid screw	M3	8	1 plan	4.50	45.00	tap drill 2.5, max 10 deep
A10	Lid screw	M3	8	1 plan	4.50	107.50	tap drill 2.5, max 10 deep
A11	Lid screw	M3	8	1 plan	4.50	155.50	tap drill 2.5, max 10 deep
A12	Lid screw	M3	8	1 plan	4.50	210.00	tap drill 2.5, max 10 deep
A13	Lid screw	M3	8	1 plan	283.50	45.00	tap drill 2.5, max 10 deep
A14	Lid screw	M3	8	1 plan	283.50	107.50	tap drill 2.5, max 10 deep
A15	Lid screw	M3	8	1 plan	283.50	155.50	tap drill 2.5, max 10 deep
A16	Lid screw	M3	8	1 plan	283.50	210.00	tap drill 2.5, max 10 deep
B1	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	42.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B2	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	69.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B3	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	96.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B4	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	163.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B5	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	201.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B6	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	239.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B7	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	249.00	16.00	boss top = board seat; tap drill 2.5, max 9 deep
B8	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	13.50	192.50	boss top = board seat; tap drill 2.5, max 9 deep
B9	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	13.50	144.50	boss top = board seat; tap drill 2.5, max 9 deep
B10	GND boss (B1..B7 Ø10, B8..B10 Ø8), joined to the wall	M3	8	1 plan	13.50	96.50	boss top = board seat; tap drill 2.5, max 9 deep
C1	Support boss S1..S6, Ø8	M3	8	1 plan	15.00	236.00	boss top = board seat; tap drill 2.5, max 9 deep
C2	Support boss S1..S6, Ø8	M3	8	1 plan	159.00	236.00	boss top = board seat; tap drill 2.5, max 9 deep
C3	Support boss S1..S6, Ø8	M3	8	1 plan	273.00	236.00	boss top = board seat; tap drill 2.5, max 9 deep
C4	Support boss S1..S6, Ø8	M3	8	1 plan	56.00	124.50	boss top = board seat; tap drill 2.5, max 9 deep
C5	Support boss S1..S6, Ø8	M3	8	1 plan	109.00	124.50	boss top = board seat; tap drill 2.5, max 9 deep
C6	Support boss S1..S6, Ø8	M3	8	1 plan	273.00	124.50	boss top = board seat; tap drill 2.5, max 9 deep
D1	Mounting, from the back	M5	THRU	1 back	258.00	230.00	tapped through floor + pad (15), exits 1 into the pocket
D2	Mounting, from the back	M5	THRU	1 back	30.00	230.00	tapped through floor + pad (15), exits 1 into the pocket
D3	Mounting, from the back	M5	THRU	1 back	258.00	35.00	tapped through floor + pad (15), exits 1 into the pocket
D4	Mounting, from the back	M5	THRU	1 back	30.00	35.00	tapped through floor + pad (15), exits 1 into the pocket
E1	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	28.00	21.00	
E2	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	55.00	21.00	
E3	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	82.00	21.00	
E4	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	149.00	21.00	
E5	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	187.00	21.00	
E6	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	225.00	21.00	
E7	SO-239 bore (Amphenol 83-798)	Ø16.7	THRU	2 wall 1	263.00	21.00	
F1	SO-239 flange screw	M3	THRU	2 wall 1	18.88	11.88	tapped 7 from outside; blind where it ends in a web
F2	SO-239 flange screw	M3	THRU	2 wall 1	37.12	11.88	tapped 7 from outside; blind where it ends in a web
F3	SO-239 flange screw	M3	THRU	2 wall 1	37.12	30.12	tapped 7 from outside; blind where it ends in a web
F4	SO-239 flange screw	M3	THRU	2 wall 1	18.88	30.12	tapped 7 from outside; blind where it ends in a web

ID	Feature	Size	Depth	View	X	Y	Note
F5	SO-239 flange screw	M3	THRU	2 wall 1	45.88	11.88	tapped 7 from outside; blind where it ends in a web
F6	SO-239 flange screw	M3	THRU	2 wall 1	64.12	11.88	tapped 7 from outside; blind where it ends in a web
F7	SO-239 flange screw	M3	THRU	2 wall 1	64.12	30.12	tapped 7 from outside; blind where it ends in a web
F8	SO-239 flange screw	M3	THRU	2 wall 1	45.88	30.12	tapped 7 from outside; blind where it ends in a web
F9	SO-239 flange screw	M3	THRU	2 wall 1	72.88	11.88	tapped 7 from outside; blind where it ends in a web
F10	SO-239 flange screw	M3	THRU	2 wall 1	91.12	11.88	tapped 7 from outside; blind where it ends in a web
F11	SO-239 flange screw	M3	THRU	2 wall 1	91.12	30.12	tapped 7 from outside; blind where it ends in a web
F12	SO-239 flange screw	M3	THRU	2 wall 1	72.88	30.12	tapped 7 from outside; blind where it ends in a web
F13	SO-239 flange screw	M3	THRU	2 wall 1	139.88	11.88	tapped 7 from outside; blind where it ends in a web
F14	SO-239 flange screw	M3	THRU	2 wall 1	158.12	11.88	tapped 7 from outside; blind where it ends in a web
F15	SO-239 flange screw	M3	THRU	2 wall 1	158.12	30.12	tapped 7 from outside; blind where it ends in a web
F16	SO-239 flange screw	M3	THRU	2 wall 1	139.88	30.12	tapped 7 from outside; blind where it ends in a web
F17	SO-239 flange screw	M3	THRU	2 wall 1	177.88	11.88	tapped 7 from outside; blind where it ends in a web
F18	SO-239 flange screw	M3	THRU	2 wall 1	196.12	11.88	tapped 7 from outside; blind where it ends in a web
F19	SO-239 flange screw	M3	THRU	2 wall 1	196.12	30.12	tapped 7 from outside; blind where it ends in a web
F20	SO-239 flange screw	M3	THRU	2 wall 1	177.88	30.12	tapped 7 from outside; blind where it ends in a web
F21	SO-239 flange screw	M3	THRU	2 wall 1	215.88	11.88	tapped 7 from outside; blind where it ends in a web
F22	SO-239 flange screw	M3	THRU	2 wall 1	234.12	11.88	tapped 7 from outside; blind where it ends in a web
F23	SO-239 flange screw	M3	THRU	2 wall 1	234.12	30.12	tapped 7 from outside; blind where it ends in a web
F24	SO-239 flange screw	M3	THRU	2 wall 1	215.88	30.12	tapped 7 from outside; blind where it ends in a web
F25	SO-239 flange screw	M3	THRU	2 wall 1	253.88	11.88	tapped 7 from outside; blind where it ends in a web
F26	SO-239 flange screw	M3	THRU	2 wall 1	272.12	11.88	tapped 7 from outside; blind where it ends in a web
F27	SO-239 flange screw	M3	THRU	2 wall 1	272.12	30.12	tapped 7 from outside; blind where it ends in a web
F28	SO-239 flange screw	M3	THRU	2 wall 1	253.88	30.12	tapped 7 from outside; blind where it ends in a web
G1	Earth strap	M5	THRU	2 wall 1	115.50	12.00	tapped through wall + boss (15)
H1	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	70.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
H2	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	118.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
H3	EXP SMA (901-9215): pocket 13 × 13 × 2 deep (corner R ≤ 1); centre bore	Ø3.0	THRU	2 wall 3	166.50	21.10	bore = 52 Ω air line around the Ø1.27 pin
J1	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	74.82	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J2	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	66.18	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J3	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	66.18	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J4	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	74.82	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J5	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	122.82	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J6	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	114.18	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J7	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	114.18	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J8	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	122.82	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J9	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	170.82	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J10	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	162.18	16.78	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J11	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	162.18	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
J12	SMA flange screw (in pocket floor)	M2	THRU	2 wall 3	170.82	25.42	upper pair tapped through (4); lower pair blind 3.5, ends 0.5 before the inner face
K1	Cutout RJ45 J1, corners R1.5	16.2 × 11.8	THRU	2 wall 2	249.50	47.50	
K2	Cutout EXP ribbon slot, corners R1.5	37 × 9	THRU	2 wall 2	214.00	30.00	
K3	Cutout XT60 J11, corners R1.5	16.5 × 9.4	THRU	2 wall 2	178.40	23.80	
K4	Cutout XT60 J12, corners R1.5	16.5 × 9.4	THRU	2 wall 2	148.40	23.80	
K5	Cutout XT30 J13, corners R1.5	11 × 6	THRU	2 wall 2	115.50	22.10	
L1	EXP plate screw	M3	THRU	2 wall 2	235.50	30.00	tapped through the wall 6
L2	EXP plate screw	M3	THRU	2 wall 2	192.50	30.00	tapped through the wall 6

X, Y: from the datum 0,0 of the view named (lower-left outer corner as drawn). Depth = full thread depth; drill depth limits in the note column.

Check: 79 of 79 tapped holes match the thread data of rf-case.solidcraft (axis, size and depth); 5 tagged on the adjacent body, geometry checked.

NOTES

1 Master geometry: rf-case.step (Tray, Lid, EXP plate). This drawing adds threads, tolerances and finish.

2 Material EN AW-6061-T6 (EN AW-6061-T6 accepted) for tray, lid and EXP plate.

3 Finish: conductive conversion coating (JLCCNC: Conductive anodizing, natural), MIL-DTL-5541 Type II Class 3 equivalent, all surfaces. No insulating anodising.

4 General tolerances ISO 2768-mK. Hole positions ±0.1 for E, F, H, J (connector flanges).

5 Threads ISO metric coarse 6H. D, G, L tapped through; F, J tapped as modelled (blind where they end in a web). Blind holes A, B, C: keep the max drill depth in the note column; never break through into the outside face.

6 Internal radii: pocket R10, every concave vertical edge inside the pocket R ≥ 6.5 (Ø12 end mill, 58 deep), cutouts R1.5. Floor edges (floor to wall, boss, pad, lug) sharp as modelled; a corner-radius end mill up to R1 is accepted.

7 Break all sharp edges 0.2 to 0.4, except the lid seat (flat, sharp, conductive).

8 Engrave the lid 0.5 deep as modelled (Arial bold).

9 Deburr the SO-239, SMA and cutout edges inside and outside (RF and cable contact).

PARTS PER UNIT

Tray 1, lid 1, EXP plate 1 (CNC, this drawing)

Printed spacer 4, printed washer 4 (sheet 5)

Screw M3 × 10 countersunk ISO 10642 A2: 16 (lid)

Screw M3 × 6 button/cheese head A2: 46 (28 SO-239, 10 GND boss, 2 RF board on S3 and S6, 4 controller board, 2 EXP plate)

Screw M5 × 16 hex socket ISO 4762 A4: 5 (4 mounting, 1 earth strap)

Screw M2 × 4 cheese head A2: 12 (SMA flanges)

Standoff M3 hex 5.5 AF, 20 mm male-female: 4 (S1, S2, S4, S5, controller)

Connector Amphenol RF 83-798 SO-239: 7. SMA Amphenol RF 901-9215: 3 (second source Cinch/Johnson 142-0701-631, D-174)

Tinned copper braid with M5 lug: 1 (earth, to the RVS earth bar)

Shack Relay Matrix rev D - RF unit case

Hole table, notes, parts list	Dwg RF-CASE-01 rev D
EN AW-6061-T6 (6082-T6 ok). Finish: conductive conversion coating (JLC 'Conductive anodizing').	Sheet 7 / 7 -
Tolerances ISO 2768-mK. Dimensions in mm. Third angle projection. Master geometry: rf-case.step	2026-11-07 PE1RGU (P. Doornekamp)